SOFTWARE TEST PLAN (STP)

Student Attendance Management System (SAMS)

Project	Student Attendance Management System (SAMS)
Document	Software Test Plan
Version	1.0
Reference	SRS v1.0, dated 05-09-2026
Date	16-09-2026
Status	Draft for review
Authors	QA Lead, Dev Lead, Test Engineer, Product Owner

1. Introduction

1.1 Purpose

This Software Test Plan (STP) defines the testing approach for the Student Attendance Management System (SAMS) v1.0. It establishes a structured, traceable process for verifying that SAMS satisfies the requirements in the Software Requirements Specification (SRS), and is the basis for test case design, test data selection, functional and security validation, defect reporting and requirement traceability.

Testing therefore focuses not only on whether individual screens work, but on whether attendance data stays correct, consistent, protected and traceable throughout the system.

Requirements use the SAMS-F-###, SAMS-NF-### and SAMS-SR-### identifiers defined by the SRS; test cases use the TC-* references from the RTM.

1.2 Scope

Testing covers end-to-end SAMS workflows for Students, Faculty, Administrators and integration services, spanning the functions in SRS Section 4, the quality requirements in Section 5 and the security requirements in Section 5.1.

1.2.1 Functional Testing Scope

- Authentication, session creation and Role-Based Access Control for Students, Faculty and Administrators.
- Administrator management of accounts, courses, sections, semesters, faculty assignments and enrollment (including bulk import).
- Attendance session creation, roster display, status recording (Present/Absent/Late/Excused), duplicate prevention and audited correction.
- Attendance calculation, role-appropriate viewing of summaries and history, and threshold-based shortage identification.
- Report generation with permitted filters, PDF/CSV export, low-attendance notifications and audit event capture.

1.2.2 Non-functional Testing Scope

- Performance: verify that 95% of normal interactive page/API requests complete within 3 seconds, excluding large report exports.
- Reliability: verify the specified monthly availability target of at least 99.5%, excluding approved maintenance windows.

- Scalability: verify support for at least 500 concurrent authenticated users and 10,000 student records in the reference deployment.
- Usability and accessibility: verify core workflows on current major desktop and mobile browsers, including keyboard navigation and readable form labels.
- Backup and recovery: verify that attendance and audit data can be backed up at least daily and restored successfully in a test environment.
- Data integrity: verify input validation, atomic attendance operations, and prevention of partial or orphaned records.

1.2.3 Security Testing Scope

- Transport security, password storage, server-side authorization, session/CSRF protection, audit logging and login-attempt throttling. Each security requirement and its acceptance criteria are specified in 5.1.3.

1.2.4 Out of Scope

The following areas are explicitly outside the scope of SAMS as defined by the SRS:

- Biometric hardware design.
- Payroll or employee attendance functionality.
- Internal university ERP functionality beyond exposed integration APIs.
- Automatic grading based on attendance.
- Special-purpose attendance hardware such as QR, RFID, biometric, or smart-card devices unless such functionality is introduced as a defined system extension.

1.3 Intended Audience

This test plan is intended for the QA Lead, Test Engineers, Developers, Product Owner or course coordinator, faculty representatives participating in UAT, project evaluators, and other stakeholders responsible for reviewing or verifying SAMS. Detailed role responsibilities are listed in 9.

1.4 References

- Student Attendance Management System — Software Requirements Specification (SRS), Version 1.0, dated 05-09-2026.
- SAMS — Software Test Plan v1.0, Sections 6, 7 and 8, dated 16-09-2026.
- SAMS Requirements Traceability Matrix (RTM).
- Relevant security and testing standards where applicable to the project.

1.5 Definitions and Test Terminology

Term	Meaning
SAMS	Student Attendance Management System.
SRS	Software Requirements Specification.
STP	Software Test Plan.
RTM	Requirements Traceability Matrix used to map requirements to test cases and verification status.
RBAC	Role-Based Access Control.
UAT	User Acceptance Testing.
Functional Requirement	A requirement describing what the system shall do.
Non-functional Requirement	A measurable quality or operational constraint such as performance, reliability, accessibility, recovery, or data integrity.
Security Requirement	A testable technical requirement that implements a security objective.
Attendance Session	A specific class meeting for which attendance is recorded.

2. Test Items

Test items are the SAMS modules, interfaces, data flows, and system behaviours selected for verification. The items below are derived from the functional requirements, non-functional requirements, security requirements, and the module mappings used in the SAMS RTM.

2.1 Authentication and Access Control Module

Requirement coverage: SAMS-F-001, SAMS-F-002.

- Validate successful login using valid credentials.
- Reject invalid credentials.
- Verify role-based access for Student, Faculty, and Administrator accounts.
- Attempt direct access to protected URLs/API endpoints without the required permission.

2.2 User Management Module

Requirement coverage: SAMS-F-003.

- Create student and faculty accounts.
- Activate and deactivate accounts.
- Verify that deactivated users cannot log in.

2.3 Course, Section and Enrollment Management Module

Requirement coverage: SAMS-F-004 and SAMS-F-005.

- Create and manage courses, sections, semesters, and faculty assignments.
- Reject invalid or duplicate faculty/course mappings where applicable.
- Enroll and remove students from sections.
- Verify supported bulk-import behaviour.

2.4 Attendance Session Module

Requirement coverage: SAMS-F-006 and SAMS-F-007.

- Create an attendance session for an assigned section.
- Verify date, start time, and optional topic/period fields.
- Reject session creation where the faculty member is not assigned to the section.
- Verify that the session roster contains active enrolled students and excludes non-enrolled students.

2.5 Attendance Recording and Correction Module

Requirement coverage: SAMS-F-008, SAMS-F-009 and SAMS-F-010.

- Mark each rostered student as Present, Absent, Late, or Excused.
- Attempt duplicate attendance insertion and verify that it is blocked or converted into an authorized update.
- Correct an existing attendance record using a reason.
- Verify that the current value changes and the old/new values, actor, time, and reason are preserved in audit history.

2.6 Attendance Calculation Module

Requirement coverage: SAMS-F-011.

- Calculate attended classes for each student and course/section.
- Calculate the attendance percentage.
- Verify that attendance summaries update after attendance changes.

2.7 Student Attendance Portal

Requirement coverage: SAMS-F-012.

- Display the student's own course-wise attendance percentage.
- Attempt to retrieve another student's attendance information.

2.8 Faculty and Administrator Reporting Module

Requirement coverage: SAMS-F-013, SAMS-F-014 and SAMS-F-015.

- Display class-wise attendance summaries.
- Identify students below the configured attendance threshold.
- Test threshold boundary conditions.
- Generate reports using permitted course/section, student, attendance-status, and date-range filters.

2.9 Report Export Module

Requirement coverage: SAMS-F-016.

- Export authorized attendance reports in CSV format.
- Export authorized attendance reports in PDF format.
- Verify generation timestamp and applied filters.

2.10 Notification Module

Requirement coverage: SAMS-F-017.

- Enable low-attendance notification behaviour where configured.
- Create a test condition in which a student's attendance crosses the configured threshold.
- Verify that one notification event is created or queued.
- Use a mail/SMS stub rather than real messages in the test environment where required.

2.11 Audit and Logging Module

Requirement coverage: SAMS-F-018 and SAMS-SR-005.

- Record failed login events.
- Record attendance creation and correction events.
- Record attendance-threshold changes.
- Verify that audit records contain timestamp, actor, action, target, outcome, and relevant metadata.

2.12 Performance, Reliability, Scalability and Recovery Items

Requirement coverage: SAMS-NF-001 through SAMS-NF-006.

- Measure the 95th percentile response time for normal interactive requests.
- Verify the monthly availability target using monitoring evidence.
- Exercise the system with at least 500 concurrent authenticated users in the reference deployment.
- Verify browser compatibility, keyboard navigation, and readable labels.
- Verify daily backup availability and successful restoration.
- Verify atomic and internally consistent attendance operations under validation and fault conditions

3. Features to be Tested

The following features of the Student Attendance Management System will be tested:

Requirement ID	Feature
SAMS-F-001	User authentication and role-based access for administrators, faculty and students
SAMS-F-002	Student registration and maintenance of student details
SAMS-F-003	Faculty/class management and assignment of students to classes
SAMS-F-004	Attendance marking for students by authorized faculty
SAMS-F-005	Attendance modification/correction by authorized users
SAMS-F-006	Attendance viewing by students and faculty
SAMS-F-007	Attendance percentage calculation and threshold validation
SAMS-F-008	Automated attendance reports generation
SAMS-F-009	Filtering reports by student, class, subject and date range
SAMS-NF-001	Report generation and attendance operations should respond within an acceptable time

4. Features Not to be Tested

The following areas are outside the scope of this test plan:

- Physical attendance infrastructure such as biometric scanners, RFID readers or other external hardware, if used.
- Third-party authentication, messaging or email services beyond verifying the application's integration with them.
- Institutional database/server hardware maintenance and operating-system administration.
- Network infrastructure and internet-service availability outside the application's control.
- Payroll, fee management, examination management and other academic modules not related to attendance.
- Internal implementation details of third-party libraries and frameworks used by the system.

5. Test Approach / Strategy

Levels:

- Unit testing – verify individual modules such as authentication, attendance calculation and report generation.
- Integration testing – verify interactions between the web application, database, authentication module and reporting components.
- System testing – verify the complete attendance workflow from login and attendance marking through calculation and report generation.
- Acceptance testing (UAT) – verify that the system satisfies the attendance-management needs of administrators, faculty and students.

Types:

- Functional testing – validate attendance marking, calculations, user roles, records and automated reports.
- Regression testing – ensure changes to attendance, user-management or reporting functionality do not break existing features.
- Performance testing – evaluate response time for attendance marking, attendance retrieval and report generation under expected user loads.
- Usability testing – verify that attendance screens, dashboards and reports are clear and easy to use.
- Security testing – verify authentication, authorization, session handling and protection of attendance records.

Entry Criteria:

- A stable application build is available.
- Required test environment, database and test accounts are ready.
- Sample student, faculty, subject and attendance data are available.
- Core attendance and reporting requirements are implemented sufficiently for testing.

Exit Criteria:

- 100% of planned test cases have been executed.
- All critical and high-severity defects are resolved or formally accepted.
- Required functional and acceptance criteria are satisfied.
- Regression testing has been completed for corrected defects and affected features.

5.1 Security Validation

Security validation verifies that SAMS protects credentials and attendance information from unauthorized disclosure, restricts access according to user roles, prevents unauthorized or untraceable changes to attendance records, and maintains sufficient evidence to identify sensitive actions and the users who performed them.

5.1.1 Security Objectives

SO-1 — Confidentiality and least privilege: Protect credentials and attendance information from unauthorized disclosure and ensure that users can access only the data required by their role.

SO-2 — Integrity and accountability: Prevent unauthorized or untraceable changes to attendance records and maintain sufficient audit evidence to identify who performed sensitive actions and when.

5.1.2 Security Validation Approach

Security validation combines positive tests, negative tests, configuration inspection, database verification, API authorization checks, session checks, and security scanning. The supplied SAMS test-plan reference identifies Postman/Newman for API testing, Selenium WebDriver for UI flows, OWASP ZAP for security scanning, and a SQL client for direct verification of password hashes and audit records.

5.1.3 Security Requirements and Validation

Req ID	Security Area	Requirement	Validation / Acceptance	Test Case
SAMS-SR-001	HTTPS/TLS	All production client-server and API communication shall use HTTPS with TLS 1.2 or higher.	Attempt HTTP access and inspect TLS configuration. HTTP shall be redirected/blocked and the security scan shall confirm TLS 1.2+ only.	TC-SEC-01
SAMS-SR-002	Password Storage	Passwords shall never be stored in plaintext and shall use the framework's approved adaptive salted password hashing mechanism.	Inspect the database and search for plaintext passwords. Stored values shall be password hashes and plaintext password searches shall return none.	TC-SEC-02
SAMS-SR-003	Server-side Authorization	Every protected page and API endpoint shall enforce authorization using the authenticated role and object-level permissions.	Attempt unauthorized page/API/object access using Student and Faculty accounts. Requests shall be denied and protected data shall not be returned.	TC-SEC-03
SAMS-SR-004	Session Security	Authenticated sessions shall use Secure/HttpOnly cookies where cookies are used, session expiration, logout invalidation, and CSRF protection for state-changing browser requests.	Inspect cookie/security configuration and execute session, logout, expiration, and CSRF tests.	TC-SEC-04
SAMS-SR-005	Audit Accountability	Sensitive administrative and attendance modifications shall create tamper-resistant audit entries containing actor, timestamp, action, target, outcome, and relevant change metadata without passwords.	Perform representative sensitive operations and inspect audit records for all required fields and absence of password data.	TC-SEC-05
SAMS-SR-006	Login Attack Protection	Repeated failed login attempts shall be rate-limited or temporarily throttled/locked according to a configurable security policy.	Automate repeated invalid-login attempts and verify that throttling or lock behaviour is triggered.	TC-SEC-06

5.1.4 Security Test Data

- Three administrator accounts and ten faculty accounts for role and authorization testing.
- Student accounts representing valid users, deactivated users, and users with different course enrollments.
- Attendance records with known values for correction and audit verification.
- Invalid credentials for authentication failure and throttling tests.
- Unauthorized role/object combinations for RBAC negative tests.

5.1.5 Security Tools and Evidence

Tool	Use	Evidence
Postman / Newman	API authentication, RBAC, and negative authorization testing.	Requests/responses and execution results.
Selenium WebDriver	Login, attendance, correction, and browser security regression flows.	Screenshots, execution logs, pass/fail results.
OWASP ZAP	TLS/session/CSRF and common security checks.	Security scan report and findings.
SQL client (psql / DBeaver)	Inspection of password hashes and audit rows.	Database inspection evidence.
Defect tracker	Security defect logging and closure.	Defect ID, severity, reproduction steps, affected requirement, and final state.

5.1.6 Security Entry and Exit Conditions

Security validation may begin when:

- A stable build is deployed to the TEST environment.
- Required security test accounts and synthetic test data are available.
- TLS/session configuration is deployed and accessible for verification.
- The security test tools and test cases are available.

Security validation is considered complete when:

- TC-SEC-01 through TC-SEC-06 have been executed.
- All High-priority security requirements have been implemented and passed.
- No Critical security defect remains open.
- Required security evidence, including scan output and database/audit verification, has been retained.
- Security requirement statuses are updated in the RTM.

5.1.7 Security Traceability

The security requirements are traceable through the RTM from the SRS requirement to the corresponding security test case. The SAMS SRS maps SAMS-SR-001 through SAMS-SR-006 to TC-SEC-01 through TC-SEC-06, allowing implementation and verification status to be tracked during testing.

6. Test Environment

SAMS is a browser-based, three-tier web application (browser client → application server → relational database). The test environment therefore has to reproduce that stack end to end, plus the optional interfaces defined in SRS Section 3.3 (institutional/ERP API, email/SMS gateway, PDF/CSV generation). No special-purpose attendance hardware is required, consistent with SRS Section 3.2.

6.1 Environment Topology

Four logically separate environments are used so that unstable code never blocks verification work:

Environment	Purpose	Notes
DEV	Developer unit testing and debugging	Local machines; SQLite or local PostgreSQL acceptable; data is disposable.
TEST (QA)	Integration, system, security and regression testing	Primary environment for this plan; mirrors production configuration with synthetic data only.
PERF	Performance, load and capacity testing (SAMS-NF-001, NF-003)	Isolated so that load runs do not distort functional test timings; seeded with 10,000 student records.
UAT	Acceptance testing by faculty/admin representatives	Frozen release candidate builds; data reset before each UAT cycle.

6.2 Hardware Requirements

Component	Specification
Application server (TEST)	Linux VM, 4 vCPU, 8 GB RAM, 50 GB SSD; HTTPS enabled with a test TLS certificate.
Database server	Linux VM, 2 vCPU, 8 GB RAM, 100 GB SSD (may be co-hosted with the app server in the mini-project deployment).
Performance load generator	Separate host, 4 vCPU, 8 GB RAM, running JMeter in non-GUI mode.
Client machines	Windows/Linux/macOS desktops with current Chrome, Firefox and Edge; at least one Android and one iOS device (or emulator) for responsive and mobile verification (SAMS-NF-004).
Backup storage	Separate volume or object-storage bucket used to verify the daily backup and restore requirement (SAMS-NF-005).
Network	Campus LAN/WAN access to the TEST server over HTTPS; ability to simulate latency and disconnection for negative-path tests.

6.3 Software Requirements

Layer	Software / Version
Application under test	SAMS v1.0 build (Django/Flask or Spring Boot as per project allocation), deployed from a tagged, versioned build artifact.
Runtime	Python 3.11+ with the chosen framework, or JDK 17+ with Spring Boot, matching the production target.
Database	PostgreSQL 14+ (or MySQL 8+) with a dedicated sams_test schema and a restorable baseline dump.
Web server / TLS	Nginx or equivalent reverse proxy configured for TLS 1.2+ only, so that SAMS-SR-001 can be verified as specified.
Operating systems	Ubuntu Server 22.04 LTS (servers); Windows 11 / Ubuntu / macOS (clients).
Browsers	Chrome, Firefox and Edge (current stable releases) plus mobile Chrome and Safari.
Mail/SMS stub	Local SMTP sink (e.g. MailHog) or a mock gateway endpoint to verify SAMS-F-017 notification events without sending real messages.
ERP/API stub	Mock REST service returning representative student, faculty, course and enrollment payloads for import testing (SAMS-F-005).

6.4 Test Tools

Tool	Category	Use in SAMS testing
pytest / JUnit	Unit testing	Module-level tests for attendance calculation, duplicate prevention and validation logic.
Postman / Newman	API testing	Authentication, RBAC and report endpoints; negative authorization cases.
Selenium WebDriver	UI automation	Login, attendance marking, correction and report-export regression flows.
Apache JMeter	Performance/load	95th-percentile response time and concurrent-user capacity.

Tool	Category	Use in SAMS testing
OWASP ZAP	Security scanning	TLS configuration, session cookie flags, CSRF and common injection checks.
SQL client (psql / DBeaver)	Data verification	Direct inspection of stored password hashes and audit rows.
Jira / GitHub Issues	Defect tracking	Defect logging, severity classification, aging and closure workflow.
Git + CI runner	Build and regression	Tagged builds delivered to TEST; automated smoke suite on every build.
Lighthouse / axe	Accessibility	Keyboard navigation and form-label checks.

6.5 Test Data

All test data is synthetic. No real student records, real names or real institutional credentials are loaded into any non-production environment, since attendance data is treated as protected under security objective SO-1.

- Users: 3 administrators, 10 faculty accounts, 200 student accounts, plus deactivated accounts for negative login tests (SAMS-F-003).
- Academic master data: 8 courses across 12 sections, with valid and deliberately conflicting faculty-assignment mappings to exercise SAMS-F-004 rejection paths.
- Enrollments: valid rosters, an unenrolled student, and a bulk-import CSV containing both valid and invalid student IDs (SAMS-F-005).
- Attendance history: a pre-seeded section with a known, hand-computed attendance percentage so that SAMS-F-011 and SAMS-F-014 results can be checked against an exact expected value.
- Threshold data: students placed just above, exactly at, and just below the configured threshold (for example 74%, 75%, 76%) to test boundary behaviour of shortage detection and notification.
- Scale dataset: 10,000 student records and roughly one academic term of attendance sessions, used only in PERF (SAMS-NF-003).
- Backup dataset: a verified database dump used to demonstrate restore for SAMS-NF-005.

6.6 Environment Setup, Access and Control

- The QA Lead owns the TEST and PERF environments; the development team owns DEV; the Product Owner (course coordinator/faculty representative) owns UAT sign-off.
- Builds are deployed to TEST only from tagged artifacts, and the build number is recorded in every execution log so defects can be tied to a specific build.

- The database is reset to the baseline dump before each full regression cycle, so results are reproducible.
- Environment configuration (framework version, DB version, TLS settings, threshold value) is recorded in a configuration sheet attached to the Test Summary Report.
- Test accounts use strong, non-shared passwords, and no environment exposes a route to production data.

6.7 Environment Assumptions and Risks

- The TEST environment is available for the full execution window; downtime beyond four hours triggers the suspension criteria of the plan.
- The ERP and email/SMS interfaces are optional in SRS Section 3.3; if the real services are unavailable, stubs are used and the corresponding test cases are marked "verified against stub" in the Test Summary Report.
- The PERF environment is sized smaller than a real production deployment; capacity results are reported as indicative for the reference deployment defined in SAMS-NF-003.

7. Test Schedule

The schedule below assumes a single-semester mini-project timeline running from mid-September to late October 2026, with test design starting as soon as the SRS baseline (v1.0, 05-09-2026) is approved. Effort is expressed in working days.

7.1 Milestones

#	Milestone / Activity	Start	End	Owner
M1	Test plan preparation and review	16-09-2026	18-09-2026	QA Lead
M2	Test case design and RTM update (all TC-* cases)	19-09-2026	25-09-2026	Test Engineer
M3	Test environment setup and smoke verification	22-09-2026	26-09-2026	QA Lead + Developer
M4	Test data preparation and seeding	24-09-2026	26-09-2026	Test Engineer
M5	Unit test completion (developer-side)	19-09-2026	30-09-2026	Developer
M6	Integration testing (app ↔ database ↔ stubs)	29-09-2026	03-10-2026	Test Engineer
M7	System testing — Cycle 1	05-10-2026	12-10-2026	Test Engineer

#	Milestone / Activity	Start	End	Owner
	(functional, all SAMS-F-###)			
M8	Security testing (SAMS-SR-001 to SR-006)	08-10-2026	13-10-2026	QA Lead
M9	Performance, reliability and recovery testing (SAMS-NF-001 to NF-005)	13-10-2026	16-10-2026	QA Lead
M10	Defect fixing and retesting	14-10-2026	20-10-2026	Developer + Test Engineer
M11	System testing — Cycle 2 (regression on fixed build)	19-10-2026	22-10-2026	Test Engineer
M12	User Acceptance Testing (UAT)	23-10-2026	27-10-2026	Faculty/Admin representatives
M13	Test Summary Report and sign-off	28-10-2026	30-10-2026	QA Lead + Product Owner

7.2 Suite-Level Execution Plan

Execution is organised by the test suites defined in SRS Section 6 (Quality Attributes & Acceptance Tests), so that every suite has an explicit window and estimated effort.

Suite	Test cases	Window	Estimated effort
Authentication & RBAC	TC-AUTH-01 to 02	05–06 Oct	1.5 person-days
Administration / master data	TC-ADM-01 to 03	06–07 Oct	2 person-days
Attendance recording	TC-ATT-01 to 05	07–09 Oct	3 person-days
Calculation & student view	TC-CALC-01, TC-STU-01	09–10 Oct	1.5 person-days
Reporting, notification, audit	TC-RPT-01 to 04, TC-NOTIF-01, TC-AUD-01	10–12 Oct	3 person-days
Security	TC-SEC-01 to 06	08–13 Oct	3 person-days
Performance &	TC-PERF-01 to 02	13–15 Oct	2 person-days

Suite	Test cases	Window	Estimated effort
scalability			
Reliability, recovery, data integrity	TC-REL-01, TC-OPS-01, TC-DATA-01	15–16 Oct	2 person-days
Usability & accessibility	TC-UX-01	16 Oct	1 person-day
Regression (Cycle 2)	All High-priority cases	19–22 Oct	3 person-days

7.3 Dependencies in the Schedule

- M7 (system testing) cannot start until M3 completes and a smoke-tested build is deployed to TEST.
- M9 (performance) depends on the PERF environment being seeded with the 10,000-record dataset at M4.
- M12 (UAT) starts only after Cycle 2 regression closes with zero open Critical or High defects.
- Test case design (M2) depends on the SRS remaining baselined; any SRS change after 25-09-2026 triggers an RTM and test case impact review, which may shift M7 onward.

7.4 Reporting Cadence

- Daily: execution status (cases planned, executed, passed, failed, blocked) during M6 to M12.
- Weekly: defect summary by severity and aging, plus requirement coverage percentage from the RTM.
- At each cycle close: a cycle report with pass rate, open defects and a recommendation on whether to proceed.
- At M13: the consolidated Test Summary Report submitted for approval.

8. Test Deliverables

The artefacts below are produced by the testing activity. Each is version-controlled alongside the SRS so that reviewers can trace a requirement from the SRS, through its test case, to its execution evidence.

8.1 Deliverables Before Test Execution

Deliverable	Description	Owner	Due
Software Test Plan (this document)	Scope, approach, environment, schedule, deliverables, risks and exit criteria for SAMS v1.0.	QA Lead	M1 — 18-09-2026

Deliverable	Description	Owner	Due
Test Case Specification	Detailed TC-* cases with preconditions, steps, test data, expected results and the requirement ID each verifies.	Test Engineer	M2 — 25-09-2026
Updated RTM	SRS Section 8 matrix extended with execution status (N/P/A) per requirement.	QA Lead	M2 — 25-09-2026
Test Data Set	Seed scripts and CSV fixtures for users, courses, enrollments, sessions and the 10,000-record scale dataset.	Test Engineer	M4 — 26-09-2026
Environment Configuration Sheet	Recorded versions, TLS settings, DB schema and threshold configuration for each environment.	QA Lead	M3 — 26-09-2026

8.2 Deliverables During Test Execution

Deliverable	Description	Owner	Frequency
Automated test scripts	Selenium UI scripts, Postman/Newman API collections, JMeter test plans and unit test suites, committed to the repository.	Test Engineer	Continuous
Test execution logs	Per-case result (Pass/Fail/Blocked) with build number, tester, timestamp and evidence (screenshots, response bodies, log excerpts).	Test Engineer	Per execution
Defect reports	Logged in the tracker with severity, priority, steps to reproduce, affected requirement ID and environment.	Test Engineer	As found
Daily status report	Cases planned vs executed, pass/fail counts, blockers and risks.	QA Lead	Daily

Deliverable	Description	Owner	Frequency
Performance test results	JMeter reports with 95th-percentile latency, throughput and error rate against SAMS-NF-001 and NF-003.	QA Lead	Per load run
Security scan reports	OWASP ZAP output and manual verification notes for SAMS-SR-001 to SR-006.	QA Lead	Per security cycle

8.3 Deliverables After Test Execution

Deliverable	Description	Owner
Test Summary Report	Consolidated results: total cases executed, pass/fail rate, defect density, defect aging, requirement coverage, open risks and an explicit statement of whether exit criteria are met.	QA Lead
Final RTM	Every SAMS-F/NF/SR requirement mapped to its test case with final status (P = Passed, A = Accepted), evidencing full coverage.	QA Lead
Defect closure report	List of all defects with final state; any deferred defect carries a documented justification and Product Owner acknowledgement.	Test Engineer
UAT sign-off record	Acceptance results and signatures from faculty/admin representatives against the acceptance criteria in SRS Section 6.	Product Owner
Backup/restore verification record	Evidence that a verified backup was restored in the test environment (SAMS-NF-005, TC-OPS-01).	QA Lead
Test archive	Scripts, data, logs and reports archived with the release tag for reproducibility and evaluation.	QA Lead

8.4 Acceptance Criteria for the Deliverables

A deliverable is considered complete when it satisfies the conditions below; these align with the acceptance exit criteria stated in SRS Section 6.

- Every requirement in the SRS has at least one test case reference in the RTM — no requirement is left unmapped.

- All NFR measurements (SAMS-NF-001 to NF-006) have recorded, quantified results rather than qualitative statements.
- The Test Summary Report is reviewed and approved by the QA Lead, Dev Lead and Product Owner.

8.5 Document Control

- All deliverables are stored in the project repository under /docs/testing/ with the version number in the filename.
- Any change after baseline is recorded in a revision history table with date, author and change summary.
- The STP, Test Case Specification and Test Summary Report are the three items submitted for mini-project evaluation.

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Joel Williams	Owns test strategy, coordinates execution, controls TEST/PERF environments, prepares consolidated reporting.
Test Engineer	Krish Gupta	Design and execute test cases, maintain test evidence, report defects.
Developer	Jaladi Suhas	Support unit testing, defect analysis, correction and retesting.
Product Owner	Kashyap V P	Participates in acceptance activities and sign-off.

10. Risks and Mitigation

Risk	Likelihood	Impact	Mitigation
Delay in stable build delivery	Medium	High	Request early smoke builds; re-sequence M7 per the dependencies in 7.3.
Test environment downtime	Medium	High	Maintain a backup cloud VM; invoke the suspension criteria in 12 beyond 4 hours.
SRS changes after baseline	Medium	Medium	Trigger an RTM and test-case impact review per 7.3; re-baseline

Risk	Likelihood	Impact	Mitigation
			affected cases.
Optional ERP / email-SMS service unavailable	High	Low	Test against stubs; mark affected cases “verified against stub” in the Test Summary Report.
PERF environment under-sized vs production	High	Medium	Report capacity results as indicative for the reference deployment (6.7); state limits in the summary.
Defect fix backlog delays Cycle 2 regression	Medium	High	Triage by severity daily; defer Low-severity defects with Product Owner sign-off per 8.4.

11. Assumptions & Dependencies

- SRS v1.0 (05-09-2026) remains the stable baseline for test design; changes after 25-09-2026 trigger the impact review described in 7.3.
- Required test data and accounts are available per the schedule in 7.
- Environment-specific assumptions and risks are detailed in 6.7.
- Security-specific assumptions are covered in 5.1.

12. Suspension & Resumption Criteria

The QA Lead decides suspension and resumption and records the decision, with its reason and duration, in the daily status report defined in 7.4. Affected test cases are marked Blocked rather than Failed so that coverage figures stay accurate.

Suspend testing if:

- The TEST environment is unavailable for more than 4 hours.

- Build instability blocks more than 30% of planned test cases.
- A Critical defect blocks a whole test suite and no workaround exists.

Resume testing if:

- Blocking defects are resolved.
- The environment is stabilized and a smoke test passes.
- The QA Lead confirms the blocking defect is fixed in a redeployed build and coverage can resume.

13. Test Case Management & Traceability

Every test case carries a unique TC-* identifier, the requirement ID it verifies, preconditions, steps, test data and an expected result. Cases are authored at M2, reviewed by the QA Lead before execution begins, and version-controlled with the SRS so a requirement can be traced forward to its evidence and back again.

The RTM is the single source of traceability and is updated at every cycle close: each requirement carries a status of N (not run), P (passed), F (failed) or A (accepted with justification). A requirement with no passing test case cannot be signed off, and any SRS change triggers the impact review in 7.3 before affected cases are re-baselined.

Module	Requirement IDs	Test cases	Status at sign-off
Authentication & RBAC	SAMS-F-001 to F-002, SAMS-SR-003	TC-AUTH-01 to 02, TC-SEC-03	P
Administration & enrollment	SAMS-F-003 to F-005	TC-ADM-01 to 03	P
Attendance recording & correction	SAMS-F-006 to F-010	TC-ATT-01 to 05	P
Calculation & student view	SAMS-F-011 to F-012	TC-CALC-01, TC-STU-01	P
Reporting, export, notification, audit	SAMS-F-013 to F-018	TC-RPT-01 to 04, TC-NOTIF-01, TC-AUD-01	P
Non-functional & security	SAMS-NF-001 to NF-006, SAMS-SR-001 to SR-006	TC-PERF, TC-REL, TC-OPS, TC-DATA, TC-UX, TC-SEC-01 to 06	P / A

14. Test Metrics & Reporting

Each metric below is computed from the execution logs and the RTM, and reported on the cadence defined in 7.4. Targets are the thresholds used to judge the exit criteria in 5 and 8.4.

Metric	Definition	Target
Test case execution	Cases executed / cases planned, per suite and overall.	100% by Cycle 2 close
Pass rate	Cases passed / cases executed.	≥ 95% at UAT entry
Defect density	Open defects / module, weighted by severity.	No module above 5 open
Defect aging	Mean days a defect stays open, by severity.	Critical ≤ 2 days, High ≤ 5 days
Requirement coverage	Requirements with at least one passed or accepted test case / total requirements in the RTM.	100% before sign-off

15. Approvals

Role	Name	Signature / Date
QA Lead	Joel Williams	Joel 16/09/2026
Dev Lead	Jaladi Suhas	Suhas 16/09/2026
Product Owner	Kashyap V P	Kashyap 16/09/2026